

HS-401: Professional Values and Ethics

LECTURE NO 5: *Normative Argumentation*

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1.1 Introduction (1/7)

Case The Pros and Cons of the Golden Gate Bridge Suicide Barrier



1.1 Introduction (2/7)

When the Golden Gate Bridge in San Francisco was finished in 1937 it was – with its span of 1280 m (4200 ft) – the world’s longest suspension bridge. It is considered to be one of the best and most beautiful examples of bridge design. The American Society of Civil Engineers has included the Golden Gate Bridge in its enumeration of the seven wonders of the modern world. However, the bridge is also the US’s most popular place to commit suicide. Since 1937, at least 1300 people killed themselves by jumping off the bridge; an average of 20 to 25 per year.¹ Suicide prevention has in fact been a concern ever since the bridge was designed. Joseph Strauss, the chief engineer is quoted as saying in 1936 (a year before the opening of the bridge): “The guard rails are five feet and six inches high [i.e. about 1.7 m] and are so constructed that any persons on the pedestrian walk could not get a handhold to climb over them. The intricate telephone and patrol systems will operate so efficiently that anyone acting suspiciously would be immediately surrounded. Suicide from the bridge is neither possible nor probable.” However, for some reason, the height of the railing was reduced in the detail design to 1.2 meter (4 feet), so making it not too difficult to climb the railing and jump off the bridge.

1.1 Introduction (3/7)

Already in 1940, the Board of Directors discussed an “anti-suicide screen” but decided against it on the basis of aesthetic and financial considerations. Another concern was that the screen might create dangerous wind resistance and make the bridge structurally unstable. Since then, proposals for a suicide barrier has been made every decade but were unsuccessful until recently. we will consider some of the main arguments of the opponents (those who are against the installation of the barrier) and the proponents (those who are in favor of installation of the barrier).

1.1 Introduction (4/7)

Aesthetic argument

- *Opponents:* The Golden Gate Bridge has been *praised for its transparency and openness*. Any barrier design would destroy the view.
- *Proponents:* Aesthetic considerations can be accounted for in the design of the barrier.

Effectiveness argument

- *Opponents:* A suicide barrier will not be *effective*, since people who want to kill themselves will simply go somewhere else.
- *Proponents:* Most suicidal people act on an impulse and, when prevented from actually killing themselves, they often seek help and lose the desire to die.

1.1 Introduction (5/7)

Economic argument

- *Opponents:* The barrier is not worth the costs. Even if the barrier would save 20 to 25 lives a year, it is not worth **50 million building costs** plus the annual operation and maintenance costs.
- *Proponents:* A human life cannot be expressed in monetary terms. “You may not think that 24 lives a year is worth spending \$50 million ... well that is until it is your daughter, son, sister, brother, friend, husband, wife etc...”

Autonomy argument

- *Opponents:* One should not interfere with **people's freedom** to commit suicide: “If I want to jump off the bridge, I don't think it is anyone's job to stop me.”
- *Proponents:* The decision to kill oneself is usually made in an impulse. As one commenter puts it: “I do believe that people have free will and that society should not interfere with their free will. But ... these people are mentally ill, and their illness can be treated. But, they cannot be treated if we let them jump off the bridge.”

1.1 Introduction (6/7)

Responsibility argument

- *Opponents:* It is not the **responsibility** of society or the directors of the bridge to try to prevent suicides. Moreover, it is wrong to use architecture to solve a social problem.
- *Proponents:* It is the responsibility of society and the bridge directors to act. As Jerome Motto, a past president of American Association of Suicidology expressed it: “If an instrument that’s being used to bring about tragic deaths is under your control, you are morally compelled to prevent its misuse.

On October 10, 2008 the Board of Directors voted 14 to 1 to install a stainless steel net which would be placed 6 m (20 ft) below the deck, and would collapse around anyone who jumped into it. The estimated costs are 50 million dollars. It is expected that construction of the net will take several years and will not start before a number of additional studies has been carried out. For more detailed information visit website: (<https://www.goldengate.org/district/district-projects/suicide-deterrent-net/>)

1.1 Introduction (7/7)



Golden Gate Bridge Suicidal Barrier

1.2 Valid Arguments (1/5)

- Before we go into the question when an argument is valid, it will be helpful to clarify some of the terms with respect to argumentation theory, that is, the study of how humans can, do, and should reach conclusions based on premises. The first term we define is *argument*, a fundamental logical concept.
- An *argument* is a set of statements, of which one (the conclusion) is claimed to follow from the others (the premises).
- *Conclusion of an argument* - The statement (the conclusion) that is affirmed on the basis of the premises of the argument.
- *Premises* - The statements, which are affirmed as providing support or reasons for accepting the conclusion.

1.2 Valid Arguments (2/5)

- In general an argument can formally be expressed as follows:
$$P_1, P_2, \dots, P_n \text{ so } C$$
- Where $P_1, P_2, \dots, P_n$ are the *premises* and C is the *conclusion*. The number of premises varies, but in practice the number will be small.
- Arguments can be judged based on their effectiveness: To what extent is an argument useful to reach the intended result? In deciding this we can distinguish two extreme analyses:
 1. The **Rhetorical analysis**: is the argument persuasive for the audience?, to be proved right is more important than to be right.
 2. The **logical analysis**: is the argument valid?
- Logic, as a discipline, is concerned with argumentation. Its concern is to distinguish a good argument from bad one, or better from worse. With that, the aim is to investigate, develop, and systematize principles and methods that can be used to distinguish valid and invalid arguments. In principle this distinction is not difficult.

1.2 Valid Arguments (3/5)

- **Valid argument**- An argument whose conclusion follows with necessity from its premises: if the premises are true, the conclusion must be true.
 - P_1 : If it rains, then streets become wet.
 - P_2 : It rains.
 - C: The streets become wet.

A statement composed of two constituent statements and the connective “if ..., then ...” is called a *conditional*. The component statement that precedes “then” is called the *antecedent*, and the component following “then” is called the *consequent*. Premise P_1 is a conditional statement, where “If it rains” is the antecedent, and “streets become wet” the consequent. For this argument it is inconceivable that the premises are true and the conclusion is false, because if the conclusion was false, then it does not rain. The argument forms an instance of the valid argument form *modus ponens*.

1.2 Valid Arguments (4/5)

- ***Modus ponens***- Form of a valid argument in which the conclusion “q” follows from the premises “p” and “if p then q.”
If p, then q
p
So, q
- ***Fallacy***- An error or deficiency in an argument.
 - P'_1 : If it rains, then I will stay at home.
 - P'_2 : I will stay at home.
 - C: It rains.
- An error or deficiency in an argument is referred to as a *fallacy*. The argument above is *invalid*. That the premises are true does not guarantee the truth of the conclusion. For example, I will stay at home since I want to watch my favourite television program. In this case, both the premises are still true, but the conclusion need not be true, which is in conflict with our definition of a valid argument.

1.2 Valid Arguments (5/5)

- Another well known valid argument form is *modus tollens*. *modus tollens* is a form of a valid argument in which the conclusion “not- p ” follows from the premises “if p then q ” and “not- q .” An instance of this form is:
 - P''_1 : If it rains, then I will stay at home.
 - P''_2 : I will not stay at home.
 - C' : It is not raining.

Modus Tollens

If p , then q

$\sim q$

So, $\sim p$

1.3 Deductive and Non-Deductive Arguments

- ***Deductive argument-*** An argument which has a conclusion that is enclosed in the premises.
- Valid arguments are *monotonic*, that is, adding new premises cannot change anything in the logical validity of a conclusion that is drawn.
- ***Non-Deductive argument-*** also known as non-monotonic arguments. Adding new premises can strengthen or weaken the conclusion.

1.4 Arguments in Ethical Theories (1 / 10)

- *1.4.1: Argumentation by analogy*
 - A type of non-deductive argumentation. An argumentation based on comparison with another situation in which the judgment is clear. The judgment is supposed also to apply to the analogous situation.

1.4 Arguments in Ethical Theories (2/10)

- *1.4.1: Argumentation by analogy*

Argumentation by Analogy

Formal description:

- Situation q is comparable with situation p (*the analogy premise*).
- If situation p occurs then r applies.
- So, if situation q occurs then r applies.

Critical questions:

- 1 Are the two situations comparable?
 - Are there important relevant similarities?
 - Are there no important relevant differences?
- 2 Is what is asserted about the example situation true? In other words, is it true that “if situation p occurs then r applies?”

1.4 Arguments in Ethical Theories (3 / 10)

- *1.4.2: Arguments in a utilitarian plea*
 - The assumption of utilitarianism to determine whether an action is acceptable, is as follows:
 - ✓ An action is morally acceptable if and only if that action can be reasonably expected to produce the greatest happiness for the greatest number of people.
- In a utilitarian plea, the *means-end argumentation* is at the forefront.
- *Means-end argumentation* is a type of non-deductive argumentation. An argumentation in which from a given end the means (actions) are derived to realize that end.

1.4 Arguments in Ethical Theories (4/10)

- *1.4.2: Arguments in a utilitarian plea*

Means-End Argumentation

If you wish to achieve end x, then you must carry out action y.

Formal:

- x (the end)
- carrying out action y (the means) realizes the end x (*the means-end premise*)
- So: do y

Critical questions:

- 1 Does action y indeed realize end x?
- 2 Can action y be carried out?
- 3 Does execution of action y lead to unacceptable side effects?
- 4 Are there no other (better) actions to achieve x?
- 5 Is the end acceptable?

1.4 Arguments in Ethical Theories (5/10)

- *1.4.2: Arguments in a utilitarian plea*
 - The most relevant critical question in a utilitarian argumentation is the first one: Does action **y** actually realize objective **x**? Two matters are of importance to answer this question. First, we must demonstrate that the action(**y**) leads to the expected consequence (**x**), and second that the consequence is the best one, that is, “the greatest happiness for the greatest number.”
 - *Causality argumentation* is a type of non-deductive argumentation. is of importance, because an expected consequence is derived from a certain action.

1.4 Arguments in Ethical Theories (6/10)

- *1.4.2: Arguments in a utilitarian plea*

Causality Argumentation

In this argumentation, we will make use of the fact that a certain expected consequence can be derived from a certain situation or action.

Formal description:

- p (action or situation)
- So: q (the expected consequence)

or

- p
- “p causes q” or “p has q as a consequence” (*the causality premise*)
- So: q

Critical questions:

- 1 Will the given situation or action indeed lead to the expected consequence?
- 2 Have no issues been forgotten, for example, with respect to the expected consequence?
- 3 How do you determine the expected consequence and can it be justified?

1.4 Arguments in Ethical Theories (7/10)

- *1.4.3: Argumentation in Kantian reasoning*
 - The assumption in Kant's theory to determine whether an action is morally acceptable is as follows:
 - ✓ An action is morally acceptable if and only if the action meets the first categorical imperative. **OR**
 - ✓ An action is morally acceptable if and only if the action meets the second categorical imperative.
 - *The first categorical imperative (the universality principle)*
 - The first categorical imperative states: You are not allowed to do anything yourself that you would not be willing to allow everyone else to do as well. You are not allowed to make exceptions for yourself. For example, if you expect other people to keep their promises, then you are obligated to keep your own promises.

1.4 Arguments in Ethical Theories (8/10)

- *1.4.3: Argumentation in Kantian reasoning*
 - *Second categorical imperative (the reciprocity principle)*
 - The second categorical imperative states: We should always respect our personal freedom and that of others to make well-considered choices.

1.4 Arguments in Ethical Theories (9/10)

- *1.4.4: Argumentation in virtue-ethical reasoning*
 - In virtue ethics, the following assumption is made to determine whether an action is morally acceptable:
 - ✓ An action is morally acceptable if and only if that action is what a virtuous agent would do in the circumstances.
 - A virtuous person exhibits good and beneficial qualities. To determine whether an action is morally acceptable, this is determined by the position someone has and what is expected of him/her in that position. Thus, a virtuous salesperson must possess the virtues of friendliness and politeness.
 - *Characteristic-judgment argumentation* is a type of non-deductive argumentation. An argument based on the assumption that a certain judgment about a thing or person can be derived from certain characteristics of that thing or person.

1.4 Arguments in Ethical Theories (10/10)

- *1.4.4: Argumentation in virtue-ethical reasoning*

Characteristic-Judgment Argumentation

If someone or something X displays certain characteristics $s_1, s_2, \dots, s_n$, then judgment A is justified for that person or thing.

Formal:

- X has the characteristics $s_1, s_2, \dots, s_n$
- characteristics $s_1, s_2, \dots, s_n$ are typical of A (*the characteristic-judgment premise*)
- So: A applies to X

Critical questions:

- 1 Do the characteristics mentioned justify judgment A?
- 2 Are the characteristics mentioned all for judgement A?
- 3 Are there any other characteristics necessary for A?
- 4 Does X possess characteristics that justify the judgment not A?
- 5 Does X possess the characteristics mentioned?

1.5 Fallacies (1/7)

- A fallacy seems to be correct but proves, on examination, not to be so. It is important to recognize fallacies, because they can mislead us. A person who recognizes fallacies is in a better position to argue successfully, and a person who cannot may fall prey to the illusions of verbal tricksters.

1.5 Fallacies (2/7)

- *1.5.1: Some common fallacies in ethical discussions*
 - There is no universally accepted classification of fallacies, but a common distinction is between *formal* and *informal fallacies*. *Formal fallacies are determined solely by the form (hence the name) or structure of an argument. Any invalid argument is a formal fallacy.* In this section we will focus on the informal fallacies. *Informal fallacy occurs because of an error in reasoning.* Informal fallacies which are going to be used in ethical discussions are: attack on the person; confusion of law and ethics; straw person; wishful thinking; naturalistic fallacy; and ambiguity.

1.5 Fallacies (3/7)

- *1.5.1: Some common fallacies in ethical discussions*

- *Attack on the person*

- ✓ This fallacy is committed when an attempt is made to discredit an argument by bringing into question in some negative way the presenter of the argument instead of attacking the argument itself.

- *Confusion of law and ethics*

- ✓ Confusion of law and ethics can simply be put as “If it isn’t illegal, it is ethical.”

- *Straw person*

- ✓ The straw person fallacy is committed when an attempt is made to misstate a person’s actual position and conclude that the original argument is a bad argument. Often this misstated argument is a distorted, exaggerated, or foolish version of the original one. Suppose that someone is critical of the use of nuclear energy, and his opponent reacts as follows: “Personally, I think the energy power supply is of the highest importance.”

1.5 Fallacies (4/7)

- *1.5.1: Some common fallacies in ethical discussions*
 - *Wishful thinking*
 - ✓ Wishful thinking (or fallacy of desire) occurs when a person interprets facts, reports, events, perceptions, and so on, according to what he/she would like to be the case rather than according to the actual or rational acceptable evidence.
 - *Naturalistic fallacy*
 - ✓ The naturalistic fallacy is the faulty assumption that everything in nature is moral by default. According to this reasoning, if something is considered being natural, it is automatically valid and justified. In the same way, any unnatural behavior is morally unacceptable. Example: Opponents of genetic modification and cloning claim that since these processes are unnatural, they are by definition undefendable and unethical.
 - *Ambiguity*
 - ✓ A person commits this fallacy when he/she uses a word or phrase unclearly. That is if that word or phrase has more than one distinct meaning (ambiguous) or has no distinct meaning (vague).

1.5 Fallacies (5/7)

- *1.5.2: Fallacies of risk*
 - In public debates on the acceptability of technological risks some specific fallacies can be identified. In this section we discuss some of these fallacies, where X and Y stand for an activity, product or technology.

1.5 Fallacies (6/7)

- *1.5.2: Fallacies of risk*

- *The sheer size fallacy*

- ✓ X is accepted
- ✓ Y is a smaller risk than X
- ✓ So, Y should be accepted

- *The fallacy of naturalness*

- ✓ X is unnatural
- ✓ So, X should not be accepted

- *The ostrich's fallacy*

- ✓ X does not give rise to any detectable risk or there is no scientific proof that X is dangerous
- ✓ So, X does not give rise to any unacceptable risk

- *The delay fallacy*

- ✓ If we wait we will know more about X
- ✓ So, no decision about X should be made now

1.5 Fallacies (7/7)

- *1.5.2: Fallacies of risk*
 - *The technocratic fallacy*
 - ✓ It is an engineering issue how dangerous X is
 - ✓ So, engineers should decide whether or not X is acceptable
 - *The fallacy of pricing*
 - ✓ We have to weight the risks of X against its benefits
 - ✓ So, we must put a price on the risks of X

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